Interpreting thyroid function tests

Lancet 2024;403:768, NICE 2023, NG145, Molecular and cellular endocrinology 2021;525:111173, JAMA 2023;329:1386, BMJ 2019;365:l2091, JAMA 2023;330:1472, American Thyroid Association www.thyroid.org, accessed June 2024

Red Whale
GEMS
Guidelines & Evidence Made Simple

The physiology

Normally:

- Thyrotropin-releasing hormone (TRH) from the hypothalamus stimulates TSH from the anterior pituitary.
- TSH acts on the thyroid gland to stimulate T4 and T3 production.
- T3 is the physiologically-active thyroid hormone. T4 is a prohormone, and is converted to T3 in peripheral tissue.
- If T3/4 rise/fall, these fluctuations are sensed by the hypothalamus and pituitary. TRH and TSH secretion is modified to buffer T3/4 via negative feedback pathways. This buffering keeps T3/T4 levels in check…usually.

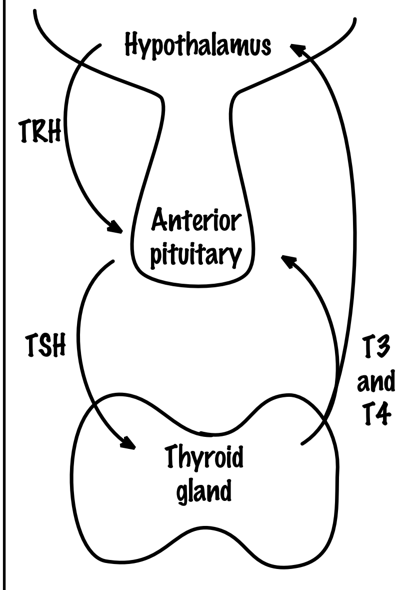
In thyroid dysfunction:

T3 and T4 production is increased/decreased, which impacts TSH output:

- In hyperthyroidism: T3/T4 production increases, and T3/4 act on the pituitary and hypothalamus, resulting in low TSH and TRH.
- In hypothyroidism: T4 production decreases, the reduction in T4 is sensed at hypothalamic and pituitary level, and TSH production increases.

In pituitary dysfunction, two patterns can occur:

- **Excess TSH production:** a TSH-producing pituitary adenoma causes excess TSH, leading to increased T3 and T4 production.
- **Compression of the anterior pituitary:** a non-TSH-producing pituitary adenoma causes pituitary compression. This reduces TSH and causes decreased T3 and T4.



TSH is the first-line test for most. If pituitary dysfunction suspected, measure TSH, T3 and T4 at the outset.

TSH
Thyroid-stimulating hormone

Raised TSH
Check T4 (usually done automatically by lab if TSH raised)

Normal TSH
A **normal** TSH usually means **normal** thyroid function …unless pituitary dysfunction is suspected…
If pituitary dysfunction suspected: check T3 andT4

Low TSH
Check T3 and T4 (usually done automatically by lab if TSH low)

Low T4 Raised TSH	Normal T4 Raised TSH	Raised T4 Raised or normal TSH	Low T3/T4 Low or normal TSH	Normal T3 and T4 Low TSH	Raised T3/T4 Low TSH
Hypothyroidism	Subclinical hypothyroidism	Consider TSH-secreting pituitary adenoma	Central hypothyroidism	Subclinical hyperthyroidism (rarely central hypothyroidism)	Hyperthyroidism
See *Hypothyroidism GEMS*	See *Subclinical hypothyroidism GEMS*	See *Pituitary adenomas* article		See *Hyperthyroidism GEMS* (or *Pituitary adenomas* article)	See *Hyperthyroidism GEMS*

When to measure which test

If the TSH is **raised** (suggesting hypothyroidism/subclinical hypothyroidism), T4 only should be checked.

- Why? Because reduction in T4 is seen before T3 starts to fall (relying on T3 here risks a delay in/missed diagnosis).

If the TSH is **low** (suggesting hyperthyroidism/subclinical hyperthyroidism), both T3 and T4 should be checked.

- Why? Because sometimes only the T3 is raised.

If central hypo/hyperthyroidism is suspected (e.g. with pituitary dysfunction), TSH, T3 and T4 should be requested.